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(54) **INFORMATION PROCESSING APPARATUS, INFORMATION PROCESSING METHOD, AND NON-TRANSITORY COMPUTER-READABLE STORAGE MEDIUM STORING PROGRAM**

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(57) **ABSTRACT**

An information processing apparatus includes a correspondence information storage portion that stores correspondence information in which a plurality of pieces of type information and a plurality of pieces of pattern information are associated with each other. The information processing apparatus includes the type information acquisition portion that acquires type information of a medium set for the print job that the printing apparatus intends to execute. The information processing apparatus includes the pattern information extraction portion that extracts pattern information corresponding to the type information acquired by the type information acquisition portion from a plurality of pieces of pattern information stored in the pattern information storage portion by referring to the correspondence information. The information processing apparatus includes the display control portion that displays the pattern information extracted by the pattern information extraction portion.

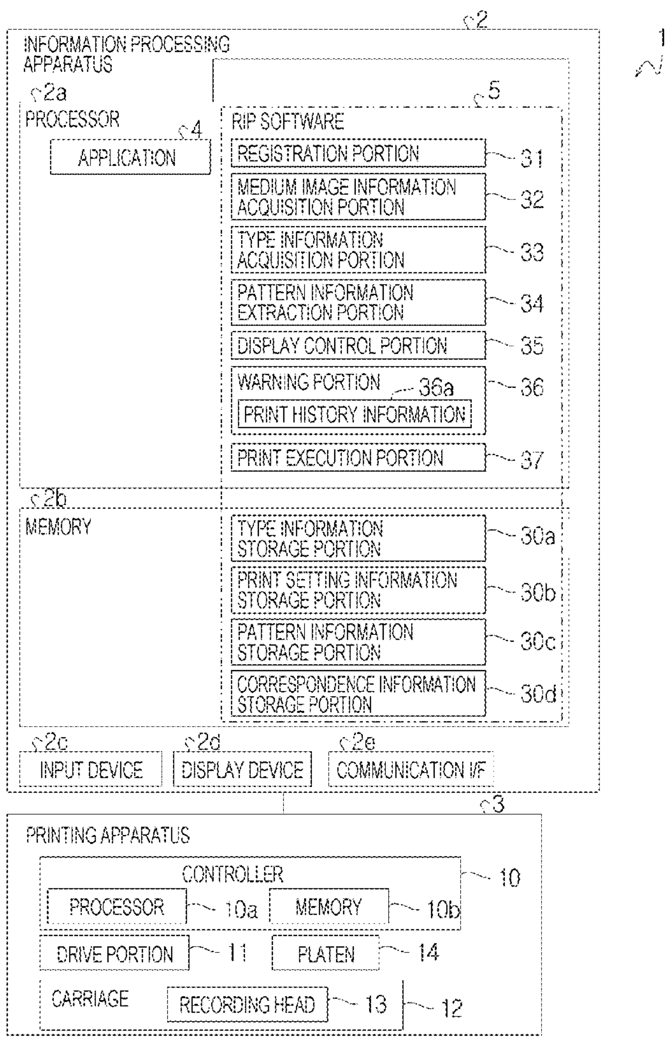
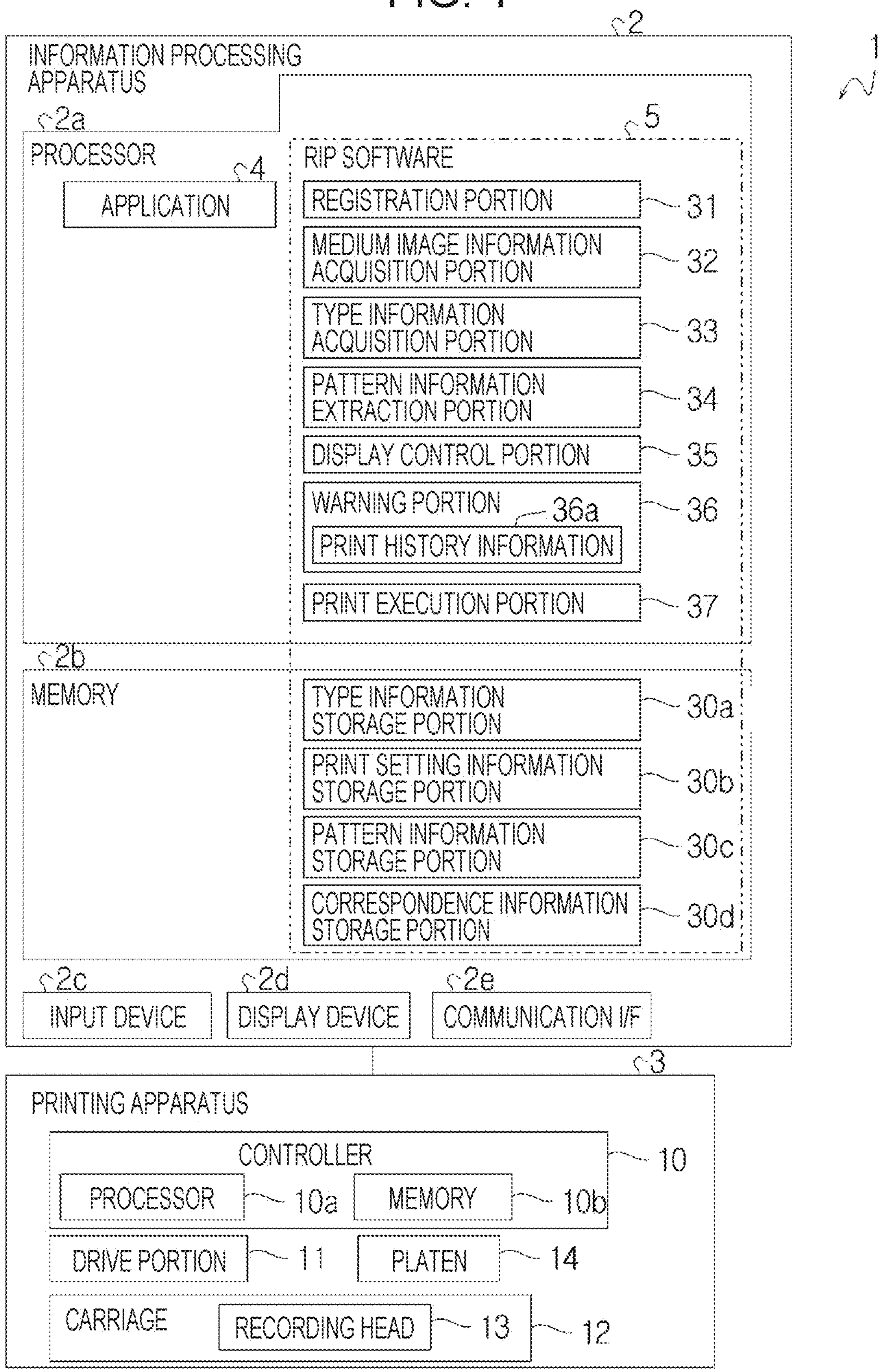


FIG. 1



2
1
INFORMATION PROCESSING APPARATUS
2a
PROCESSOR
4
APPLICATION
5
RIP SOFTWARE
REGISTRATION PORTION
31
MEDIUM IMAGE INFORMATION ACQUISITION PORTION
32
TYPE INFORMATION ACQUISITION PORTION
33
PATTERN INFORMATION EXTRACTION PORTION
34
DISPLAY CONTROL PORTION
35
WARNING PORTION
36a
PRINT HISTORY INFORMATION
36
PRINT EXECUTION PORTION
37
2b
MEMORY
TYPE INFORMATION STORAGE PORTION
30a
PRINT SETTING INFORMATION STORAGE PORTION
30b
PATTERN INFORMATION STORAGE PORTION
30c
CORRESPONDENCE INFORMATION STORAGE PORTION
30d
2c
INPUT DEVICE
2d
DISPLAY DEVICE
2e
COMMUNICATION I/F
3
PRINTING APPARATUS
CONTROLLER
10
PROCESSOR
10a
MEMORY
10b
DRIVE PORTION
11
PLATEN
14
CARRIAGE
RECORDING HEAD
13
12

FIG. 2

MEDIUM TYPE	PRINT SETTING			POSITIONING PATTERN
	COLOR CONVERSION PROFILE	RECORDING METHOD 1	RECORDING METHOD 2	
smt_m_1	profile_70	BASE WHITE/ABSENT	RECIPROCATING	smt_p_1
smt_m_2	profile_91	BASE WHITE/PRESENT	RECIPROCATING	smt_p_2
smt_m_3	profile_62	BASE WHITE/ABSENT	ONLY OUTBOUND	smt_p_3
smt_m_4	profile_30	BASE WHITE/ABSENT	RECIPROCATING	smt_p_4
smt_m_5	profile_75	BASE WHITE/ABSENT	RECIPROCATING	smt_p_5
smt_m_6	profile_85	BASE WHITE/ABSENT	RECIPROCATING	smt_p_6
badge_m_1	profile_64	BASE WHITE/ABSENT	ONLY OUTBOUND	badge_p_1
badge_m_2	profile_27	BASE WHITE/ABSENT	ONLY OUTBOUND	badge_p_2
keyho_m_1	profile_93	BASE WHITE/PRESENT	RECIPROCATING	keyho_p_1
keyho_m_2	profile_73	BASE WHITE/PRESENT	RECIPROCATING	keyho_p_2
keyho_m_3	profile_18	BASE WHITE/ABSENT	RECIPROCATING	keyho_p_3
keyho_m_4	profile_37	BASE WHITE/PRESENT	RECIPROCATING	keyho_p_4
keyho_m_5	profile_66	BASE WHITE/PRESENT	RECIPROCATING	keyho_p_5

FIG. 3

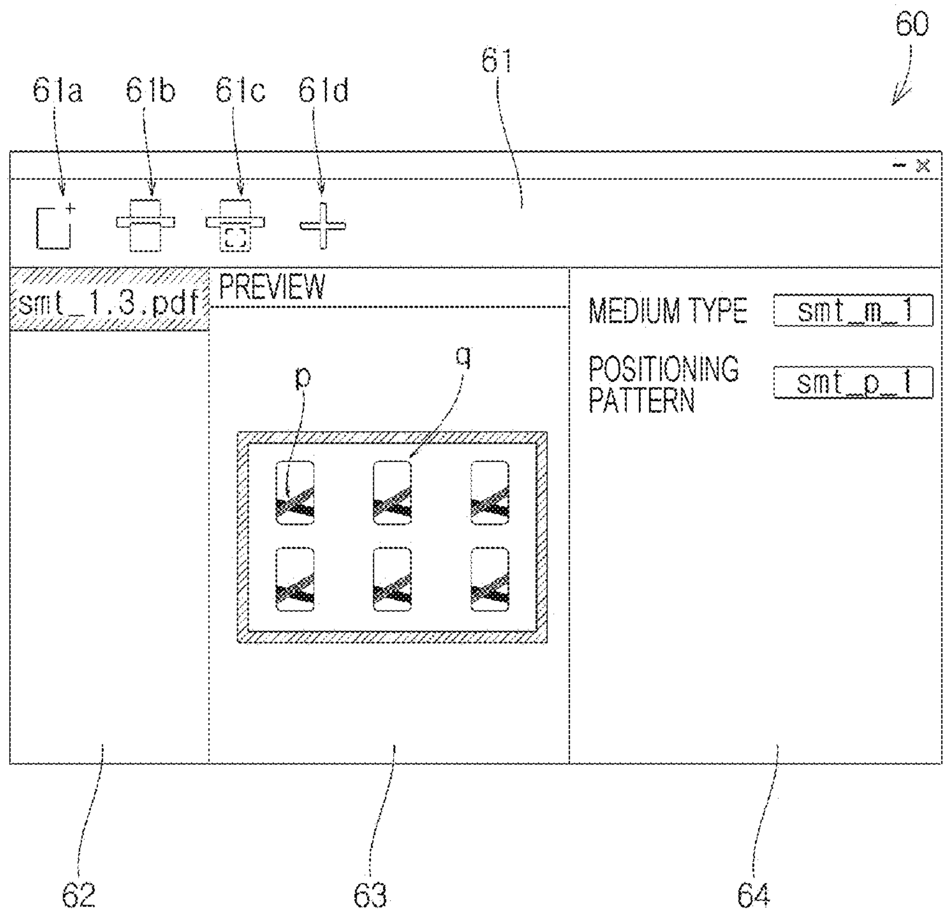

60
61a
61b
61c
61d
61
smt_1.3.pdf
PREVIEW
p
q
MEDIUM TYPE
smt_m_1
POSITIONING PATTERN
smt_p_1
62
63
64

FIG. 4

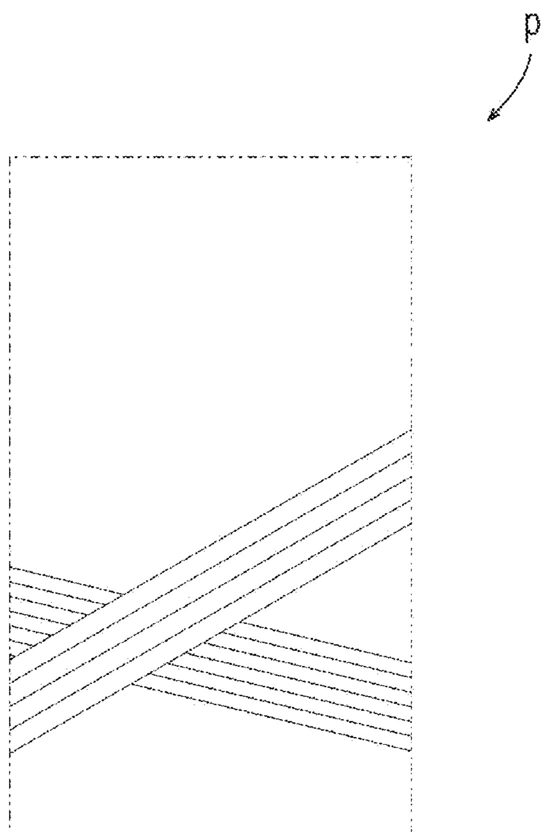


FIG. 5

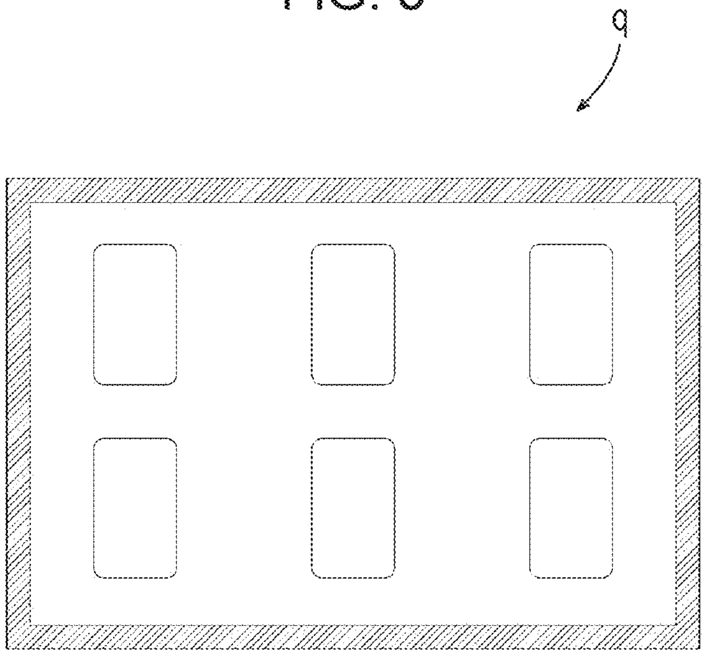


FIG. 6

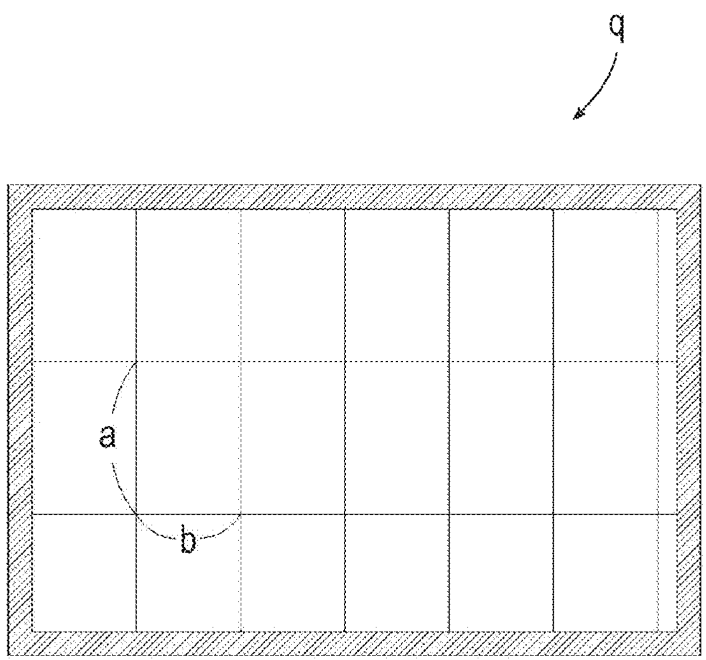


FIG. 7

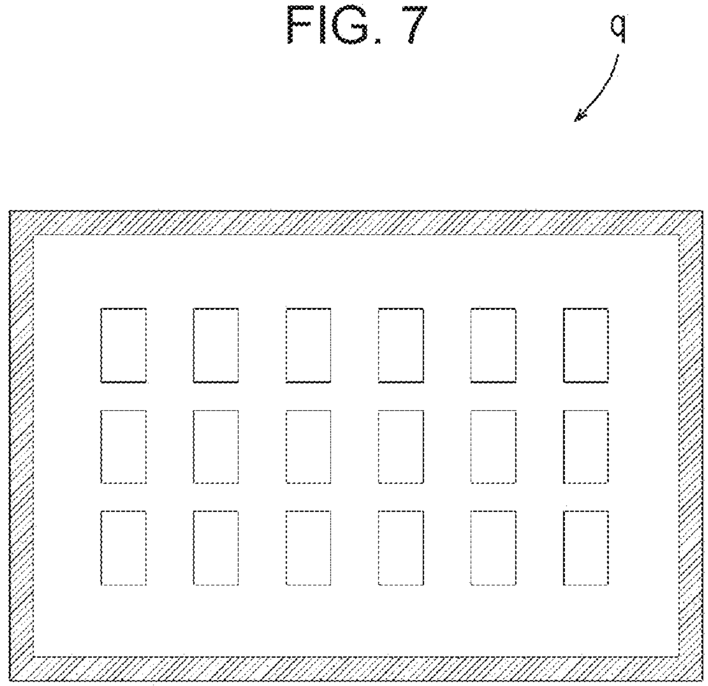


FIG. 8

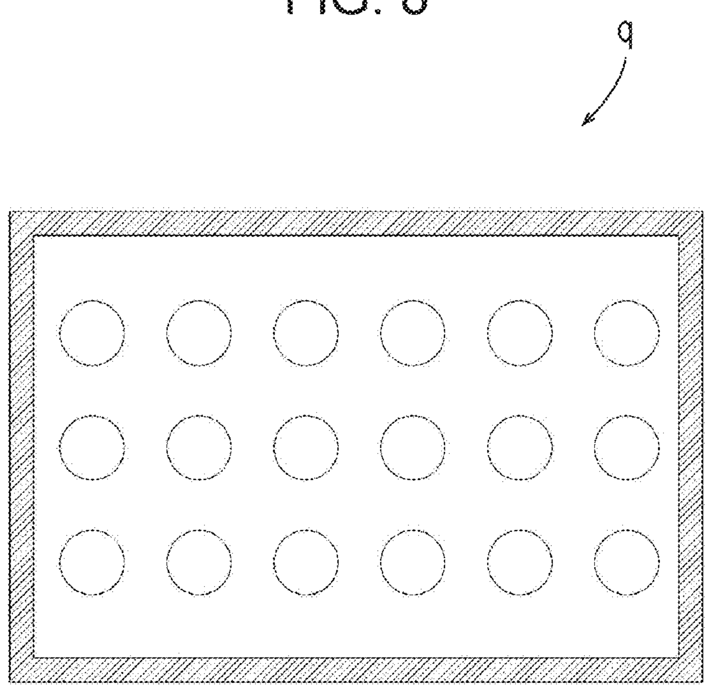


FIG. 9

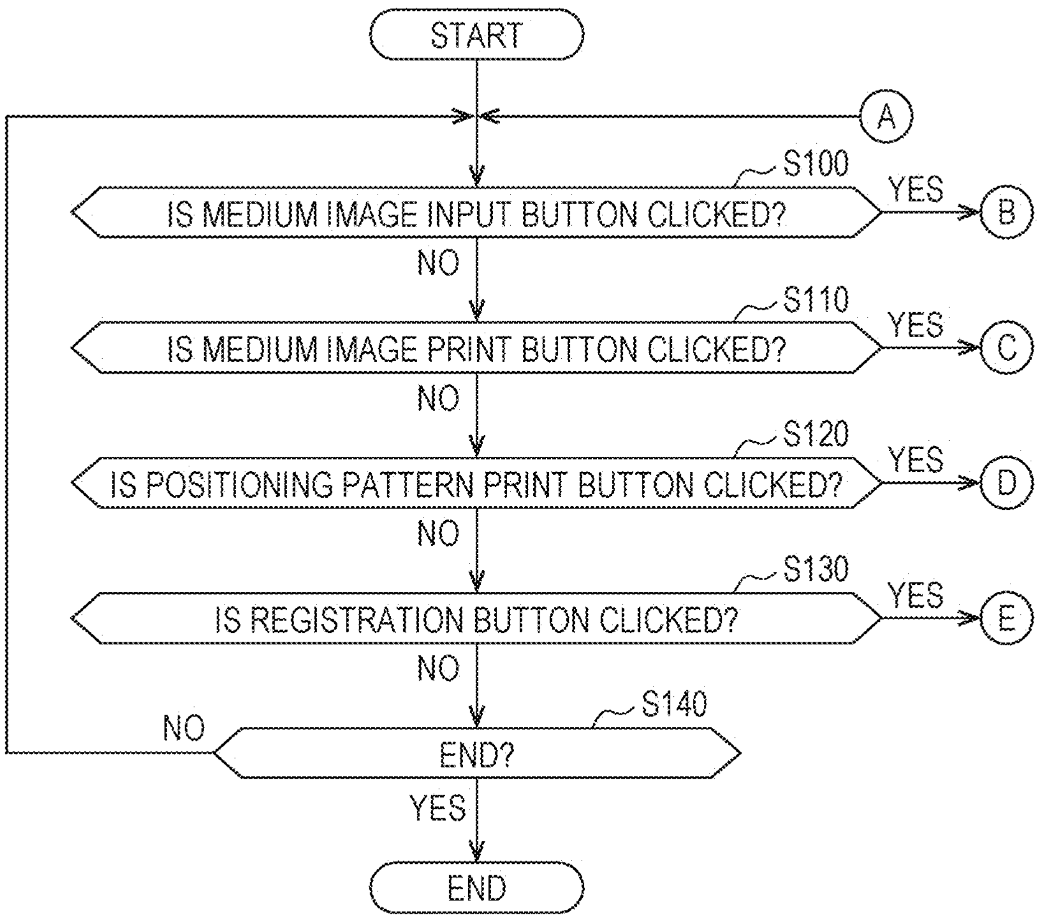

START
A
S100
IS MEDIUM IMAGE INPUT BUTTON CLICKED?
YES
B
NO
S110
IS MEDIUM IMAGE PRINT BUTTON CLICKED?
YES
C
NO
S120
IS POSITIONING PATTERN PRINT BUTTON CLICKED?
YES
D
NO
S130
IS REGISTRATION BUTTON CLICKED?
YES
E
NO
S140
NO
END?
YES
END

FIG. 10

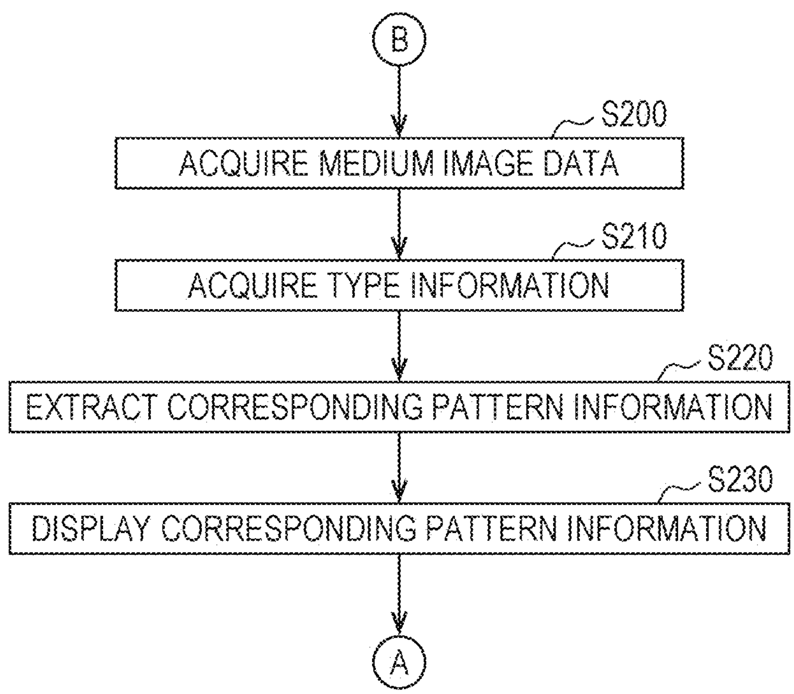

B
S200
ACQUIRE MEDIUM IMAGE DATA
S210
ACQUIRE TYPE INFORMATION
S220
EXTRACT CORRESPONDING PATTERN INFORMATION
S230
DISPLAY CORRESPONDING PATTERN INFORMATION
A

FIG. 11

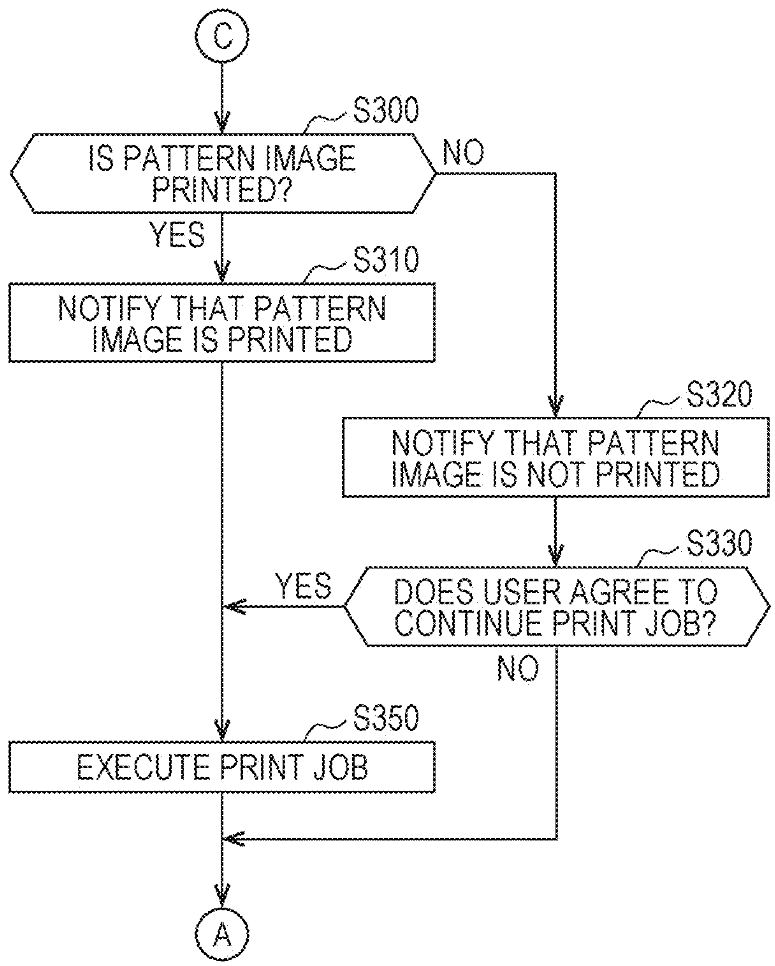

C
S300
IS PATTERN IMAGE PRINTED?
NO
YES
S310
NOTIFY THAT PATTERN IMAGE IS PRINTED
S320
NOTIFY THAT PATTERN IMAGE IS NOT PRINTED
S330
YES
DOES USER AGREE TO CONTINUE PRINT JOB?
NO
S350
EXECUTE PRINT JOB
A

FIG. 12

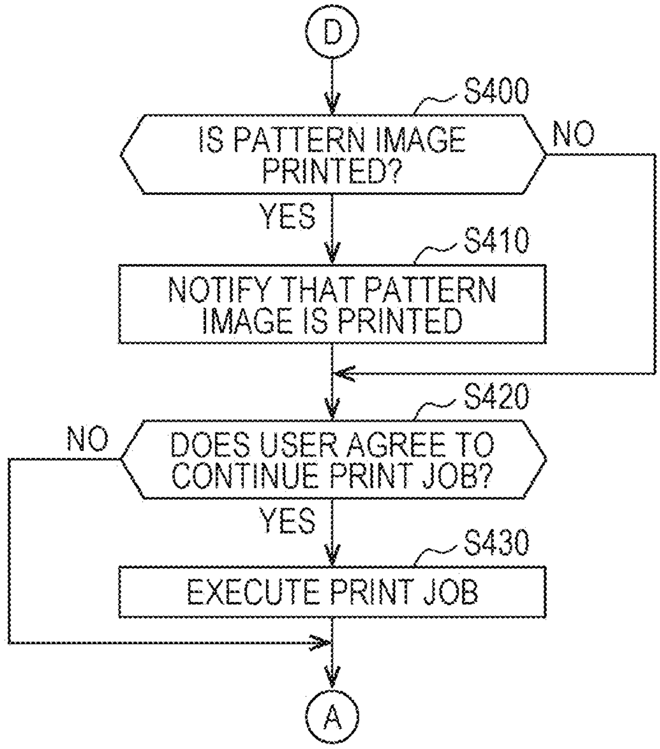

D
S400
IS PATTERN IMAGE PRINTED?
NO
YES
S410
NOTIFY THAT PATTERN IMAGE IS PRINTED
S420
NO
DOES USER AGREE TO CONTINUE PRINT JOB?
YES
S430
EXECUTE PRINT JOB
A

FIG. 13

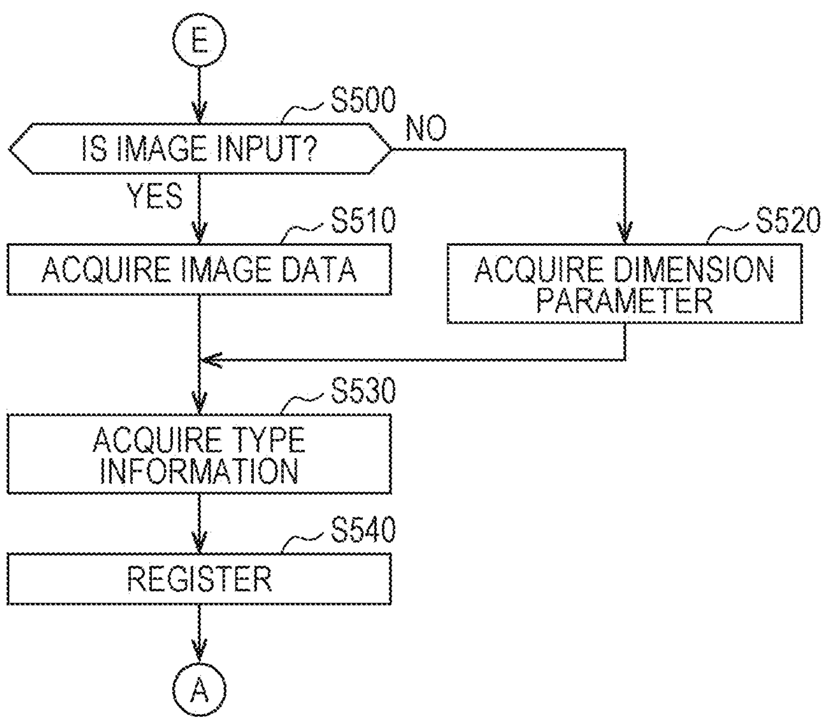

E
S500
IS IMAGE INPUT?
NO
YES
S510
ACQUIRE IMAGE DATA
S520
ACQUIRE DIMENSION PARAMETER
S530
ACQUIRE TYPE INFORMATION
S540
REGISTER
A

FIG. 14

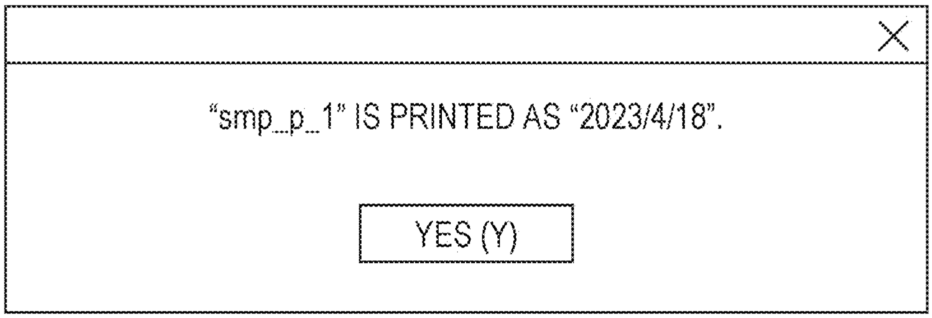


FIG. 15

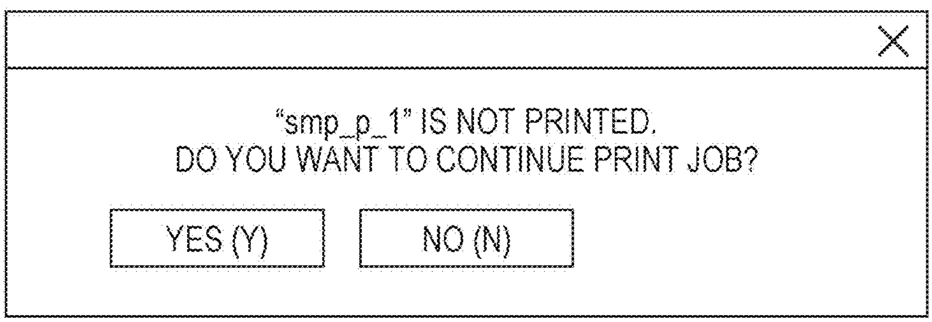
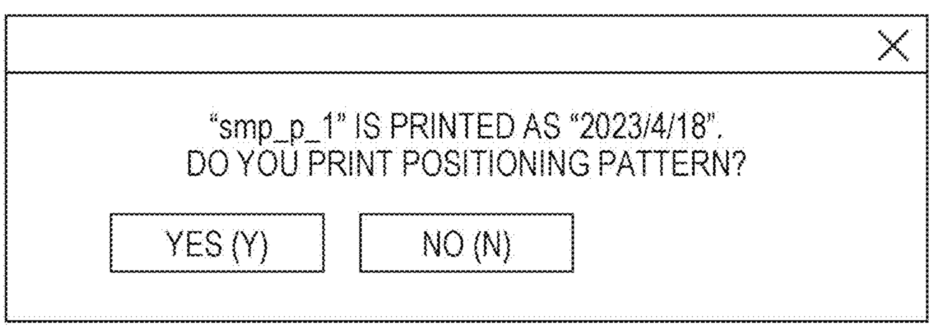


FIG. 16



INFORMATION PROCESSING APPARATUS, INFORMATION PROCESSING METHOD, AND NON-TRANSITORY COMPUTER-READABLE STORAGE MEDIUM STORING PROGRAM

[0001] The present application is based on, and claims priority from JP Application Serial Number 2024-032764, filed Mar. 5, 2024, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to an information processing apparatus, an information processing method, and a non-transitory computer-readable storage medium storing a program.

2. Related Art

[0003] JP-A-2022-159645 discloses that, when printing on a first medium using a flatbed printer, a second medium for positioning the first medium is generated and printed in order to dispose the first medium at an appropriate position of a placement table of the flatbed printer. The image printed on the second medium for positioning includes a frame line indicating the contour of the first medium.

[0004] In JP-A-2022-159645, when performing printing on the first medium, the first medium is positioned on the placement table using the second medium different from the second medium corresponding to the first medium, and as a result, there is a concern that the first medium may be lost.

SUMMARY

[0005] According to an aspect of the present disclosure, there is an information processing apparatus that controls a printing apparatus executing a print job on a print medium positioned by using a positioning medium on which a positioning pattern is printed, the information processing apparatus including: a type information storage portion that stores a plurality of pieces of type information each indicating a plurality of different types of the print medium; a pattern information storage portion that stores a plurality of pieces of pattern information each indicating a plurality of the positioning patterns different from each other; a correspondence information storage portion that stores correspondence information for associating the plurality of pieces of type information and the plurality of pieces of pattern information with each other; a type information acquisition portion that acquires type information of the print medium set for the print job that the printing apparatus intends to execute; a pattern information extraction portion that extracts, from the plurality of pieces of pattern information, pattern information corresponding to the type information acquired by the type information acquisition portion by referring to the correspondence information; and a pattern information display portion that displays the pattern information extracted by the pattern information extraction portion.

[0006] According to another aspect of the present disclosure, there is provided an information processing method for controlling a printing apparatus executing a print job on a print medium positioned by using a positioning medium on which a positioning pattern is printed, in which a computer stores a plurality of pieces of type information each indicating a plurality of different types of the print medium, stores a plurality of pieces of pattern information each indicating a plurality of the positioning patterns different from each other, stores correspondence information for associating the plurality of pieces of type information and the plurality of pieces of pattern information with each other, acquires type information of the print medium set for the print job that the printing apparatus intends to execute, extracts, from the plurality of pieces of pattern information, pattern information corresponding to the acquired type information by referring to the correspondence information, and displays the extracted pattern information.

[0007] According to still another aspect of the present disclosure, there is provided a non-transitory computer-readable storage medium storing a program that causes a computer that controls a printing apparatus executing a print job on a print medium positioned by using a positioning medium on which a positioning pattern is printed, to function as a type information storage portion that stores a plurality of pieces of type information each indicating a plurality of different types of the print medium, a pattern information storage portion that stores a plurality of pieces of pattern information each indicating a plurality of the positioning patterns different from each other, a correspondence information storage portion that stores correspondence information for associating the plurality of pieces of type information and the plurality of pieces of pattern information with each other, a type information acquisition portion that acquires type information of the print medium set for the print job that the printing apparatus intends to execute, a pattern information extraction portion that extracts, from the plurality of pieces of pattern information, pattern information corresponding to the type information acquired by the type information acquisition portion by referring to the correspondence information, and a pattern information display portion that displays the pattern information extracted by the pattern information extraction portion.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. **1** is a block diagram of a printing system.

[0009] FIG. **2** is a data structure diagram of correspondence information.

[0010] FIG. **3** is a diagram illustrating a GUI of RIP software.

[0011] FIG. **4** is a diagram illustrating an example of a medium image.

[0012] FIG. **5** is a diagram illustrating an example of a positioning pattern.

[0013] FIG. **6** is a diagram illustrating an example of a positioning pattern.

[0014] FIG. **7** is a diagram illustrating an example of a positioning pattern.

[0015] FIG. **8** is a diagram illustrating an example of a positioning pattern.

[0016] FIG. **9** is a control flow when the GUI is displayed.

[0017] FIG. **10** is a control flow when a medium image input button is clicked.

[0018] FIG. **11** is a control flow when a medium image print button is clicked.

[0019] FIG. **12** is a control flow when a positioning pattern print button is clicked.

[0020] FIG. **13** is a control flow when a registration button is clicked.

[0021] FIG. **14** is a diagram illustrating a display example of a dialog box.

[0022] FIG. **15** is a diagram illustrating a display example of a dialog box.

[0023] FIG. **16** is a diagram illustrating a display example of a dialog box.

DESCRIPTION OF EMBODIMENTS

[0024] Hereinafter, the present disclosure will be described through embodiments of the present disclosure, but the present disclosure is not limited to the following embodiments. In addition, not all of the configurations described in the embodiment are essential as means that solves the problem. For clarification of the description, the following description and drawings are appropriately omitted and simplified. In each drawing, the same elements will be given the same reference numerals, and the redundant description thereof will be omitted as necessary.

Summary of Printing System 1

[0025] FIG. **1** is a block diagram of a printing system **1**. As illustrated in FIG. **1**, the printing system **1** includes an information processing apparatus **2** and a printing apparatus **3**. The information processing apparatus **2** is typically a personal computer. The printing apparatus **3** is a flatbed printer.

[0026] The information processing apparatus **2** includes a processor **2***a*, a memory **2***b*, an input device **2***c*, a display device **2***d*, and a communication interface **2***e*. The information processing apparatus **2** communicates with the printing apparatus **3** via the communication interface **2***e*. The processor **2***a* can access the memory **2***b*. The processor **2***a* reads and executes a program stored in the memory **2***b*. As a result, the program causes hardware such as the processor **2***a* to execute the functions of an application **4** and a raster image processor (RIP) software **5**.

[0027] The input device **2***c* is a unit for receiving an operation by a user. The input device **2***c* is typically realized by a keyboard or a pointing device.

[0028] The display device **2***d* is a unit for displaying visual information. The display device **2***d* is typically realized by a liquid crystal display (LCD) or an organic light emitting diode (OLED).

[0029] The application **4** generates medium image data indicating a medium image to be printed on a medium in accordance with a drawing instruction from the user. The application **4** outputs the medium image data to the RIP software **5** in accordance with a print instruction from the user.

[0030] The medium is a specific example of a print medium. The medium has a certain thickness, and is typically a smartphone case, a badge, or a key holder, but is not limited to these types. These media are sometimes called commercial products or media, and for example, there are a huge number of types, even just for smartphone cases. In addition, the material may be different when the type of the medium is different, and accordingly, it is necessary to change the print setting parameter when the printing is performed by the printing apparatus **3**.

[0031] The RIP software **5** generates first print control data based on the medium image data, and transmits the generated first print control data to the printing apparatus **3**. The first print control data is data for the printing apparatus **3** to print a medium image.

[0032] In addition, the RIP software **5** generates second print control data based on the pattern information indicating the positioning pattern for disposing the medium at an appropriate position in the printing apparatus **3**, and transmits the generated second print control data to the printing apparatus **3**. The second print control data is data for the printing apparatus **3** to print the positioning pattern. The RIP software **5** is also referred to as a printer driver. Details of the RIP software **5** will be described later.

[0033] The print medium on which the positioning pattern is printed is typically paper. Therefore, the paper on which the positioning pattern is printed is also referred to as a positioning paper below. The positioning paper is a specific example of a positioning medium.

[0034] The printing apparatus **3** includes a controller **10**, a drive portion **11**, a carriage **12**, a recording head **13**, a platen **14**, and an ultraviolet (UV) irradiation portion (not illustrated).

[0035] The controller **10** includes a processor **10***a* and a memory **10***b*. The processor **10***a* executes processing according to predetermined firmware stored in the memory **10***b*.

[0036] The platen **14** is typically a flat plate in a horizontal posture. A plurality of media are arranged on the platen **14** when printing is simultaneously performed on the plurality of media. Therefore, it can be said that the platen **14** supports a plurality of media. When simultaneously printing on a plurality of media, typically, a positioning paper is set on the platen **14**, and the plurality of media are positioned with respect to the platen **14** by using a positioning pattern printed on the positioning paper. In the present embodiment, the platen **14** is provided with the heater for the purpose of promoting the drying of the ink, and the platen temperature can be adjusted by the heater.

[0037] The recording head **13** discharges a liquid such as ink by an ink jet method to perform recording. The recording head **13** includes a plurality of nozzles and discharges ink to a plurality of media or papers arranged on the platen **14**. The controller **10** controls the recording head **13** with a predetermined discharge waveform based on the first print control data or the second print control data received from the information processing apparatus **2**, and discharges or does not discharge the ink from the plurality of nozzles. As a result, the recording head **13** applies the ink to the medium at a desired ink density. In addition, the ink applied on the medium forms an ink layer, and a plurality of ink layers may be provided.

[0038] The carriage **12** and the drive portion **11** relatively move the recording head **13** with respect to the platen **14** and the medium. The carriage **12** is provided with the recording head **13**, and reciprocates the recording head **13** in the main scanning direction with respect to the platen **14** and the medium. The speed of this reciprocating movement is referred to as a head speed. The drive portion **11** moves the recording head **13** in the sub-scanning direction intersecting the main scanning direction with respect to the platen **14** and the medium. Further, the drive portion **11** changes the distance between the platen **14** and the nozzles of the recording head **13** in the vertical direction intersecting the main scanning direction and the sub-scanning direction. This distance is also referred to as a platen gap. The controller **10** changes the scanning speed, which is the moving speed of the carriage **12**, or the size of the platen gap according to the print setting.

[0039] The UV irradiation portion is mounted on, for example, the carriage **12**, and moves in the main scanning direction with respect to the platen **14** and the medium, similar to the recording head **13**. At this time, the ink on the medium is solidified or cured by irradiating the ink on the medium with ultraviolet rays. The controller **10** varies the UV irradiation intensity or the UV irradiation frequency by the UV irradiation portion according to the print setting.

Details of RIP Software **5**

[0040] The processor **2*a*** functions as a registration portion **31**, a medium image information acquisition portion **32**, a type information acquisition portion **33**, a pattern information extraction portion **34**, a display control portion **35**, a warning portion **36**, and a print execution portion **37** by executing the program of the RIP software **5**. Further, the memory **2*b*** functions as a type information storage portion **30*a***, a print setting information storage portion **30*b***, a pattern information storage portion **30*c***, and a correspondence information storage portion **30*d*** by storing various information in accordance with the program of the RIP software **5**.

[0041] The type information storage portion **30*a*** stores a plurality of pieces of type information. Each of the plurality of pieces of type information indicates a plurality of different types of media. Each piece of type information typically includes an identification character string for identifying the corresponding type. The identification character string for identifying the type is also referred to as a type identification character string below.

[0042] The print setting information storage portion **30*b*** stores a plurality of pieces of print setting information. Each of the plurality of pieces of print setting information indicates a plurality of different print settings. Each print setting information is typically configured with a plurality of print setting parameters.

[0043] The pattern information storage portion **30*c*** stores a plurality of pieces of pattern information. Each of the plurality of pieces of pattern information indicates a plurality of different positioning patterns. Each piece of pattern information may include image information of the corresponding positioning pattern. In this case, each piece of pattern information includes an identification character string for identifying the corresponding positioning pattern, and the image information. Each pattern information may include dimension parameters of the grid lines when the corresponding positioning pattern is a grid line that assists in positioning the corresponding medium. In this case, each pattern information includes an identification character string for identifying the corresponding positioning pattern, and the dimension parameter. The identification character string for identifying the positioning pattern is also referred to as a pattern identification character string below.

[0044] The correspondence information storage portion **30*d*** associates the plurality of pieces of type information, the plurality of pieces of print setting information, and the plurality of pieces of parameter information with each other. FIG. **2** illustrates a data structure diagram of the correspondence information. In FIG. **2**, the "medium type" corresponds to the type information. The character strings such as "smt_m_1", "smt_m_2", and "keyho_m_1" illustrated in "medium type" are type identification character strings of the corresponding type information. The "print setting" corresponds to the print setting information. The "positioning pattern" corresponds to the pattern information. The character strings such as "smt_p_1", "smt_p_2", and "keyho_p_1" illustrated in the "positioning pattern" are pattern identification character strings of the corresponding pattern information.

[0045] "smt_m_1" to "smt_m_6" are all type identification character strings related to the smartphone case.

[0046] "badge_m_1" and "badge_m_2" are both type identification character strings related to the badge.

[0047] "keyho_m_1" to "keyho_m_5" are all type identification character strings related to the key holder.

[0048] In the present embodiment, as illustrated in FIG. **2**, the correspondence information associates one piece of pattern information with one piece of type information in a one-to-one manner. However, instead of this, a plurality of pieces of pattern information may be associated with one piece of type information in a one-to-many manner.

[0049] In the present embodiment, the print setting information storage portion **30*b*** can be omitted. Therefore, the description of the print settings will be omitted below.

[0050] The registration portion **31** registers new correspondence relationship in the correspondence information in the correspondence information storage portion **30*d***. Specifically, the registration portion **31** acquires the type information and the pattern information. The registration portion **31** registers the acquired type information in the type information storage portion **30*a*** and registers the pattern information in the pattern information storage portion **30*c***. Then, the registration portion **31** registers the correspondence relationship between the type information and the pattern information in the correspondence information storage portion **30*d***.

[0051] The medium image information acquisition portion **32** acquires the medium image data.

[0052] The type information acquisition portion **33** acquires type information of the medium set for the print job to be printed by the printing apparatus **3**.

[0053] The pattern information extraction portion **34** extracts at least one piece of pattern information corresponding to the type information of the medium acquired by the type information acquisition portion **33** from the plurality of pieces of pattern information stored in the pattern information storage portion **30*c*** by referring to the correspondence information of the correspondence information storage portion **30*d***.

[0054] The display control portion **35** is a specific example of a pattern information display portion and a trigger display portion.

[0055] The display control portion **35** causes the display device **2*d*** to display the pattern information extracted by the pattern information extraction portion **34** in a character format. Specifically, the display control portion **35** causes the display device **2*d*** to display a pattern identification character string of the pattern information extracted by the pattern information extraction portion **34**.

[0056] Further, the display control portion **35** causes the display device **2*d*** to display the positioning pattern indicated by the pattern information extracted by the pattern information extraction portion **34** in an image format. Specifically, the display control portion **35** causes the display device **2*d*** to display an image (hereinafter also referred to as a pattern image) of the positioning pattern indicated by the pattern information extracted by the pattern information extraction portion **34**. Here, the "image of the positioning pattern" refers to a case where the pattern information includes the image information of the corresponding positioning pattern. In this case, the display control portion **35** displays the image information as a pattern image on the display device **2***d*. On the other hand, when the pattern information includes the dimension parameter of the corresponding positioning pattern, the display control portion **35** generates a pattern image based on the dimension parameter and displays the generated pattern image on the display device **2***d*.

[0057] Further, the display control portion **35** simultaneously displays the first trigger for starting printing on the medium and the second trigger for starting printing of the pattern image displayed by the display control portion **35**, on the display device **2***d*. Specifically, the following is the case.

[0058] With reference to FIG. **3**, a graphical user interface (GUI) **60** of the RIP software **5** to be displayed on the display device **2***d* by the display control portion **35** will be described. The GUI **60** includes a toolbar **61**, a file name list box **62**, a preview box **63**, and a print job panel **64**.

[0059] The display control portion **35** displays, in a clickable button format, a medium image input button **61***a* for inputting medium image data to the toolbar **61**, a medium image print button **61***b* as a first trigger, a positioning pattern print button **61***c* as a second trigger, and a registration button **61***d* for registering the type information and the pattern information in association with the correspondence information of the correspondence information storage portion **30***d*.

[0060] The display control portion **35** displays the file name of the medium image data imported by the user to the RIP software **5** in the file name list box **62**. In other words, the display control portion **35** displays the file name of the medium image data set for the print job in the file name list box **62**.

[0061] The display control portion **35** displays a medium image p and a pattern image q indicated by the medium image data in the preview box **63** in a superimposed manner.

[0062] Here, refer to FIGS. **4** to **8**. FIG. **4** illustrates an example of the medium image p. In FIG. **4**, a two-dot chain line indicates an outer edge of the medium image p. FIGS. **5** to **8** illustrate the pattern image q of the positioning pattern. The positioning pattern illustrated in FIG. **5** is a positioning pattern for a smartphone case and includes an outer contour line of the smartphone case. The positioning pattern illustrated in FIG. **6** is also a positioning pattern for a smartphone case and is configured with grid lines. The positioning pattern illustrated in FIG. **7** is a positioning pattern for a key holder and includes an outer contour line of the key holder. The positioning pattern illustrated in FIG. **8** is a positioning pattern for a badge and includes an outer contour line of the badge.

[0063] The display control portion **35** displays the print job setting on the print job panel **64**. The print job setting includes at least the type information and the pattern information. That is, the display control portion **35** displays both the type identification character string of the type information acquired by the type information acquisition portion **33** and the pattern identification character string of the pattern information extracted by the pattern information extraction portion **34**. In addition, when a plurality of pieces of pattern information correspond to the type information displayed on the print job panel **64**, the display control portion **35** may display the pattern identification character string in a pull-down format, and accordingly, the user can change the pattern information of the print job setting. In the example illustrated in FIG. **3**, "smt_m_1" is set as the type information and "smt_p_1" is set as the pattern information in the print job.

[0064] The warning portion **36** issues a warning to the user by displaying a dialog box. Specifically, the warning portion **36** holds print history information **36***a*. The print history information **36***a* indicates the print history of the pattern image corresponding to each pattern information illustrated in FIG. **2**. The warning portion **36** refers to the print history information **36***a* and issues a warning to the user. For example, when the medium image print button **61***b* is clicked despite the fact that the pattern image corresponding to the pattern information set in the print job was not printed even once during a predetermined past period, or when the positioning pattern print button **61***c* is clicked despite the fact that the pattern image corresponding to the pattern information set for the print job was printed during the predetermined past period, the warning portion **36** issues a warning to the user. A specific warning aspect by the warning portion **36** will be described later.

[0065] The print execution portion **37** executes a print job. Specifically, when the medium image print button **61***b* is clicked, the print execution portion **37** converts the medium image data set for the print job into the first print control data, and transmits the first print control data to the printing apparatus **3**. As a result, the printing apparatus **3** performs printing based on the first print control data. Further, when the positioning pattern print button **61***c* is clicked, the print execution portion **37** generates the second print control data based on the pattern information set for the print job, and transmits the generated second print control data to the printing apparatus **3**. As a result, the printing apparatus **3** performs printing based on the second print control data.

[0066] Next, a control flow of the RIP software **5** will be described with reference to FIGS. **9** to **13**. FIG. **9** is a control flow when the GUI **60** is displayed. FIG. **10** is a control flow when the medium image input button **61***a* is clicked. FIG. **11** is a control flow when the medium image print button **61***b* is clicked. FIG. **12** is a control flow when the positioning pattern print button **61***c* is clicked. FIG. **13** is a control flow when the registration button **61***d* is clicked.

[0067] As illustrated in FIG. **9**, the RIP software **5** determines whether the medium image input button **61***a* is clicked (S**100**). When it is determined that the medium image input button **61***a* is clicked (S**100**: YES), the RIP software **5** proceeds the process to step S**200** in FIG. **10**. When it is determined that the medium image input button **61***a* is not clicked (S**100**: NO), the RIP software **5** proceeds the process to step S**110**.

[0068] The RIP software **5** determines whether the medium image print button **61***b* is clicked (S**110**). When it is determined that the medium image print button **61***b* is clicked (S**110**: YES), the RIP software **5** proceeds the process to step S**300** in FIG. **11**. When it is determined that the medium image print button **61***b* is not clicked (S**110**: NO), the RIP software **5** proceeds the process to step S**120**.

[0069] The RIP software **5** determines whether the positioning pattern print button **61***c* is clicked (S**120**). When it is determined that the positioning pattern print button **61***c* is clicked (S**120**: YES), the RIP software **5** proceeds the process to step S**400** in FIG. **12**. When it is determined that the positioning pattern print button **61***c* is not clicked (S**120**: NO), the RIP software **5** proceeds the process to step S**130**.

[0070] The RIP software **5** determines whether the registration button **61***d* is clicked (S**130**). When it is determined that the registration button **61***d* is clicked (S**130**: YES), the RIP software **5** proceeds the process to step S**500** in FIG. **13**. When it is determined that the registration button **61***d* is not clicked (S**130**: NO), the RIP software **5** proceeds the process to step S**140**.

[0071] The RIP software **5** determines whether the GUI **60** is closed (S**140**). When it is determined that the GUI **60** is closed (S**140**: YES), the RIP software **5** ends the process. When it is not determined that the GUI **60** is closed (S**140**: NO), the RIP software **5** returns the process to step S**100**.

When Medium Image Input Button **61***a* Is Clicked

[0072] As illustrated in FIG. **10**, the medium image information acquisition portion **32** of the RIP software **5** displays an open file dialog. As a result, the medium image information acquisition portion **32** acquires the medium image data selected by the user via the open file dialog (S**200**). The medium image information acquisition portion **32** sets the medium image data selected by the user as the medium image data for the print job.

[0073] Next, the type information acquisition portion **33** of the RIP software **5** acquires the type information by analyzing the medium image data selected by the user or based on the input operation of the user (S**210**). The type information acquisition portion **33** sets the type information acquired by the type information acquisition portion **33** as the type information in the print job.

[0074] Next, the pattern information extraction portion **34** of the RIP software **5** extracts the pattern information corresponding to the type information acquired by the type information acquisition portion **33** by referring to the correspondence information (S**220**), and sets the pattern information as the pattern information for the print job.

[0075] Next, the display control portion **35** displays the pattern information extracted by the pattern information extraction portion **34** to the user (S**230**), and returns the process to step S**100** in FIG. **9**. Specifically, the display control portion **35** displays the pattern information extracted by the pattern information extraction portion **34** on the GUI **60** in a character format and an image format. As a result, the user can appropriately understand the positioning pattern corresponding to the medium to be printed in the future by checking the display content on the GUI **60**, and thus, a mismatch between the medium and the positioning pattern can be prevented.

When Medium Image Print Button **61***b* Is Clicked

[0076] As illustrated in FIG. **11**, the warning portion **36** determines whether the pattern image corresponding to the pattern information set for the print job is printed within the first period from the time when the medium image print button **61***b* is clicked by referring to the print history information **36***a* (S**300**).

[0077] When it is determined that the pattern image corresponding to the pattern information set for the print job is printed within the first period from the time when the medium image print button **61***b* is clicked (S**300**: YES), as illustrated in FIG. **14**, the warning portion **36** notifies the user that the pattern image corresponding to the pattern information set for the print job is printed by using a dialog box (S**310**), and proceeds the process to step S**350**.

[0078] When it is determined that the pattern image corresponding to the pattern information set for the print job is not printed within the first period from the time when the medium image print button **61***b* is clicked (S**300**: NO), as illustrated in FIG. **15**, the warning portion **36** notifies the user that the pattern image corresponding to the pattern information set for the print job is not printed by using a dialog box (S**320**), and prompts the user to agree to continue the print job (S**330**). When the user agrees in step S**330** (S**330**: YES), the warning portion **36** proceeds the process to step S**350**. When the user does not agree in step S**330** (S**330**: NO), the warning portion **36** returns the process to step S**100** in FIG. **9**.

[0079] In step S**350**, the print execution portion **37** converts the medium image data into the first print control data, transmits the first print control data to the printing apparatus **3** (S**350**), and returns the process to step S**100** in FIG. **9**. As a result, the printing apparatus **3** executes printing on the medium according to the first print control data.

[0080] The above-described first period is typically set to about one month or three months, but the present disclosure is not limited thereto.

When Positioning Pattern Print Button **61***c* Is Clicked

[0081] As illustrated in FIG. **12**, the warning portion **36** determines whether the pattern image corresponding to the pattern information set for the print job is printed within the second period from the time when the positioning pattern print button **61***c* is clicked by referring to the print history information **36***a* (S**400**).

[0082] When it is determined that the pattern image corresponding to the pattern information set for the print job is printed within the second period from the time when the positioning pattern print button **61***c* is clicked (S**400**: YES), as illustrated in FIG. **16**, the warning portion **36** notifies the user that the pattern image corresponding to the pattern information set for the print job is printed by using a dialog box (S**410**), and prompts the user to agree to continue printing the positioning pattern set for the print job (S**420**). When the user agrees in step S**420** (S**420**: YES), the warning portion **36** proceeds the process to step S**430**. On the other hand, when the user does not agree in step S**420** (S**420**: NO), the warning portion **36** returns the process to step S**100** in FIG. **9**.

[0083] When it is determined that the pattern image corresponding to the pattern information set for the print job is not printed within the second period from the time when the positioning pattern print button **61***c* is clicked (S**400**: NO), the warning portion **36** proceeds the process to step S**430**.

[0084] In step S**430**, the print execution portion **37** converts the pattern image corresponding to the pattern information set for the print job into the second print control data, transmits the second print control data to the printing apparatus **3** (S**430**), and returns the process to step S**100** in FIG. **9**. As a result, the printing apparatus **3** executes printing on the paper according to the second print control data.

[0085] The above-described second period is typically set to about one month or three months, but the present disclosure is not limited thereto. The second period may have the same time width as the first period, or may be longer or shorter than the first period.

When Registration Button **61***d* Is Clicked

[0086] As illustrated in FIG. **13**, when registering pattern information in the correspondence information, the registration portion **31** asks the user in a dialog box format whether to input the pattern information in an image format (S**500**). When the user selects to input the pattern information in an image format (S**500**: YES), the registration portion **31** displays the open file dialog. As a result, the registration portion **31** acquires the image data selected by the user via the open file dialog (S**510**). On the other hand, when the user does not select to input the positioning pattern in an image format (S**500**: NO), the registration portion **31** estimates that the pattern information to be registered in the correspondence information by the user is configured with the dimension parameter of the grid line, and displays the input screen for inputting the dimension parameter of the grid line. Here, the "dimension parameter of the grid line" includes at least a vertical distance a and a horizontal distance b of the grid line as illustrated in FIG. **6**. As a result, the registration portion **31** acquires the dimension parameter input by the user via the input screen (S**520**).

[0087] Next, the registration portion **31** acquires the type information based on the user input (S**530**).

[0088] Next, the registration portion **31** registers the type information acquired in step S**530** in the type information storage portion **30***a*. The registration portion **31** registers the pattern information acquired in step S**510** or step S**520** in the pattern information storage portion **30***c*. The registration portion **31** registers the type information acquired in step S**530** and the pattern information acquired in step S**510** or step S**520** in association with each other in the correspondence information (S**540**), and returns the process to step S**100** in FIG. **9**.

[0089] In the above, a preferred embodiment of the present disclosure was described. The above-described embodiment has the following features.

[0090] That is, the information processing apparatus **2** controls the printing apparatus **3** that executes the print job for the medium positioned by using the positioning paper on which the positioning pattern is printed. The information processing apparatus **2** includes the type information storage portion **30***a* that stores a plurality of pieces of type information indicating a plurality of types different from each other in the medium. The information processing apparatus **2** includes a pattern information storage portion **30***c* that stores a plurality of pieces of pattern information indicating a plurality of positioning patterns different from each other. The information processing apparatus **2** includes the correspondence information storage portion **30***d* that stores correspondence information in which a plurality of pieces of type information and a plurality of pieces of pattern information are associated with each other. The information processing apparatus **2** includes the type information acquisition portion **33** that acquires type information of a medium set for the print job that the printing apparatus **3** intends to execute. The information processing apparatus **2** includes the pattern information extraction portion **34** that extracts pattern information corresponding to the type information acquired by the type information acquisition portion **33** from a plurality of pieces of pattern information stored in the pattern information storage portion **30***c* by referring to the correspondence information. The information processing apparatus **2** includes the display control portion **35** that displays the pattern information extracted by the pattern information extraction portion **34**. According to the above configuration, the positioning pattern suitable for the print job that the printing apparatus **3** intends to execute is displayed, and thus a mismatch between the medium and the positioning pattern can be prevented.

[0091] Further, the display control portion **35** displays the pattern information in a character format. According to the above configuration, the user can appropriately understand the positioning pattern suitable for the print job by viewing the pattern identification character string displayed on the print job panel **64**.

[0092] Further, the display control portion **35** displays the pattern information in an image format. According to the above configuration, the user can appropriately understand the positioning pattern suitable for the print job by viewing the pattern image q displayed in the preview box **63**.

[0093] Further, the display control portion **35** displays the pattern image q (image) of the positioning pattern indicated by the pattern information and the medium image p (image) to be printed on the medium in a superimposed manner. According to the above configuration, the user can check the consistency between the medium image p and the pattern image q by simultaneously viewing the medium image p and the pattern image q displayed in the preview box **63**.

[0094] In addition, the pattern information may be image information indicating an outer contour line of the corresponding medium. The pattern information may be dimension parameters of the grid line that assists in positioning the corresponding medium.

[0095] The information processing apparatus **2** further includes the registration portion **31** that acquires the type information and the pattern information and registers the correspondence relationship between the type information and the pattern information in correspondence information. According to the above configuration, new media or new positioning patterns can be easily registered in the correspondence information.

[0096] Further, the display control portion **35** simultaneously displays the medium image print button **61***b* for starting printing on the medium and the positioning pattern print button **61***c* for starting printing of the pattern information displayed by the display control portion **35**. According to the above configuration, printing of the positioning pattern can be executed as necessary before the printing on the medium is started.

[0097] Further, when the pattern information displayed by the display control portion **35** is not printed within the first period from the time when the medium image print button **61***b* is operated, the warning portion **36** issues a warning by notification as illustrated in FIG. **15**. According to the above configuration, it is estimated that there is no appropriate positioning paper, and thus printing on the medium can be stopped.

[0098] Further, when the pattern information displayed by the display control portion **35** is printed within the second period from the time when the positioning pattern print button **61***c* is operated, the warning portion **36** issues a warning by notification as illustrated in FIG. **16**. According to the above configuration, it is estimated that an appropriate positioning paper is already present, and thus the same positioning pattern can be effectively prevented from being printed a plurality of times.

[0099] The above embodiment can be modified as follows, for example.

[0100] That is, the information processing apparatus **2** may share the print history of each positioning pattern indicated by each pattern information with another information processing apparatus, and may appropriately refer to the shared print history. That is, when one printing apparatus **3** is shared by the plurality of information processing apparatuses **2**, there is a concern that the positioning paper may be created in duplicate when each information processing apparatus **2** individually holds the print history of each positioning pattern. Therefore, by sharing the print history among the plurality of information processing apparatuses **2**, the positioning paper can be effectively prevented from being created in duplicate.

[0101] In the above example, the program can be stored by using various types of non-transitory computer readable media and can be supplied to the computer. The non-transitory computer readable medium includes various types of tangible storage media. Examples of non-transitory computer readable media include magnetic recording media (for example, a flexible disk, a magnetic tape, and a hard disk drive) and magneto-optical recording media (for example, a magneto-optical disk). Examples of non-transitory computer readable media include read only memory (CD-ROM), CD-R, CD-R/W, and semiconductor memory (for example, mask ROM). Examples of non-transitory computer readable media further include programmable ROM (PROM), erasable PROM (EPROM), flash ROM, and random access memory (RAM). The program may be supplied to the computer by various types of transitory computer readable media. Examples of the temporary computer readable medium include an electric signal, an optical signal, and an electromagnetic wave. The temporary computer readable medium can supply a program to a computer via a wired communication path such as a wire and an optical fiber or a wireless communication path.

What is claimed is:

1. An information processing apparatus that controls a printing apparatus executing a print job on a print medium positioned by using a positioning medium on which a positioning pattern is printed, the information processing apparatus comprising:

a type information storage portion that stores a plurality of pieces of type information each indicating a plurality of different types of the print medium;

a pattern information storage portion that stores a plurality of pieces of pattern information each indicating a plurality of the positioning patterns different from each other;

a correspondence information storage portion that stores correspondence information for associating the plurality of pieces of type information and the plurality of pieces of pattern information with each other;

a type information acquisition portion that acquires type information of the print medium set for the print job that the printing apparatus intends to execute;

a pattern information extraction portion that extracts, from the plurality of pieces of pattern information, pattern information corresponding to the type information acquired by the type information acquisition portion by referring to the correspondence information; and

a pattern information display portion that displays the pattern information extracted by the pattern information extraction portion.

2. The information processing apparatus according to claim **1**, wherein

the pattern information display portion displays the pattern information in a character format.

3. The information processing apparatus according to claim **1**, wherein

the pattern information display portion displays the pattern information in an image format.

4. The information processing apparatus according to claim **3**, wherein

the pattern information display portion displays an image of the positioning pattern indicated by the pattern information and an image printed on the print medium in a superimposed manner.

5. The information processing apparatus according to claim **1**, wherein

the pattern information is image information indicating an outer contour line of the corresponding print medium.

6. The information processing apparatus according to claim **1**, wherein

the pattern information is a dimension parameter of a grid line that assists in positioning of the corresponding print medium.

7. The information processing apparatus according to claim **1**, further comprising:

a registration portion that acquires the type information and the pattern information and registers a correspondence relationship between the type information and the pattern information in the correspondence information.

8. The information processing apparatus according to claim **1**, further comprising:

a trigger display portion that simultaneously displays a first trigger for starting printing on the print medium and a second trigger for starting printing of the positioning pattern indicated by the pattern information displayed by the pattern information display portion.

9. The information processing apparatus according to claim **1**, further comprising:

a trigger display portion that displays a first trigger for starting printing on the print medium; and

a warning portion that issues a warning when the positioning pattern indicated by the pattern information is not printed within a first period from a time when the first trigger is operated.

10. The information processing apparatus according to claim **1**, further comprising:

a trigger display portion that displays a second trigger for starting printing of the positioning pattern indicated by the pattern information displayed by the pattern information display portion; and

a warning portion that issues a warning when the positioning pattern indicated by the pattern information is printed within a second period from a time when the second trigger is operated.

11. The information processing apparatus according to claim **9**, wherein the information processing apparatus shares a print history of each positioning pattern indicated by each pattern information with another information processing apparatus.

12. An information processing method for controlling a printing apparatus executing a print job on a print medium positioned by using a positioning medium on which a positioning pattern is printed, wherein

a computer

stores a plurality of pieces of type information each indicating a plurality of different types of the print medium,

stores a plurality of pieces of pattern information each indicating a plurality of the positioning patterns different from each other,

stores correspondence information for associating the plurality of pieces of type information and the plurality of pieces of pattern information with each other,

acquires type information of the print medium set for the print job that the printing apparatus intends to execute,

extracts, from the plurality of pieces of pattern information, pattern information corresponding to the acquired type information by referring to the correspondence information, and

displays the extracted pattern information.

13. A non-transitory computer-readable storage medium storing a program, the program that causes a computer that controls a printing apparatus executing a print job on a print medium positioned by using a positioning medium on which a positioning pattern is printed, to function as

a type information storage portion that stores a plurality of pieces of type information each indicating a plurality of different types of the print medium,

a pattern information storage portion that stores a plurality of pieces of pattern information each indicating a plurality of the positioning patterns different from each other,

a correspondence information storage portion that stores correspondence information for associating the plurality of pieces of type information and the plurality of pieces of pattern information with each other,

a type information acquisition portion that acquires type information of the print medium set for the print job that the printing apparatus intends to execute,

a pattern information extraction portion that extracts, from the plurality of pieces of pattern information, pattern information corresponding to the type information acquired by the type information acquisition portion by referring to the correspondence information, and

a pattern information display portion that displays the pattern information extracted by the pattern information extraction portion.

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